



National Load Despatch Centre
राष्ट्रीय भार प्रेषण केंद्र
GRID CONTROLLER OF INDIA LIMITED
ग्रिड कंट्रोलर ऑफ इंडिया लिमिटेड

(Government of India Enterprise/ भारत सरकार का उद्यम)
B-9, QUTUB INSTITUTIONAL AREA, KATWARIA SARAI, NEW DELHI -110016
बी-9, कुतुब इन्स्टीट्यूशनल एरिया, कटवारिया सराये, न्यू दिल्ली-110016

Ref: POSOCO/NLDC/SO/Daily PSP Report

दिनांक: 21.09.2024

To,

- कार्यकारी निदेशक, पू.क्षे.भा.प्रे.के., 14, गोल्फ क्लब रोड, कोलकाता - 700033
Executive Director, ERLDC, 14 Golf Club Road, Tollygunge, Kolkata, 700033
- कार्यकारी निदेशक, ऊ.क्षे.भा.प्रे.के., 18/ए, शाहीद जीत सिंह सनसनवाल मार्ग, नई दिल्ली - 110016
Executive Director, NRLDC, 18-A, Shaheed Jeet Singh Marg, Katwaria Sarai, New Delhi - 110016
- कार्यकारी निदेशक, प.क्षे.भा.प्रे.के., एफ3, एम आई डी सी क्षेत्र, अंधेरी, मुंबई - 400093
Executive Director, WRLDC, F-3, M.I.D.C. Area, Marol, Andheri (East), Mumbai-400093
- कार्यकारी निदेशक, ऊ.पू.क्षे.भा.प्रे.के., डोंगतेह, लोअर नोंग्रह, लापालंग, शिलोंग - 793006
Executive Director, NERLDC, Dongteih, Lower Nongrah, Lapalang, Shillong - 793006, Meghalaya
- कार्यकारी निदेशक, द.क्षे.भा.प्रे.के., 29, रेस कोर्स क्रॉस रोड, बंगलुरु - 560009
Executive Director, SRLDC, 29, Race Course Cross Road, Bangalore-560009

Sub: Daily PSP Report for the date 20.09.2024.

महोदय/Dear Sir,

आईईजीसी-2023 की धारा स.-5.5.1 के प्रावधान के अनुसार, दिनांक 20-सितंबर-2024 की अखिल भारतीय प्रणाली की दैनिक ग्रिड निष्पादन रिपोर्ट रांभांप्रेके की वेबसाइट पर उपलब्ध है।

As per article 5.5.1 of the Indian Electricity Grid Code, the daily report pertaining power supply position of All India Power System for the date 20.09.2024, is available at the NLDC website.

धन्यवाद,
Thanks

Report for previous day Date of Reporting: 21-Sep-2024

A. Power Supply Position at All India and Regional level

	NR	WR	SR	ER	NER	TOTAL
Demand Met during Evening Peak hrs(MW) (at 20:00 hrs; from RLDCs)	68266	62866	49761	29134	3753	213780
Peak Shortage (MW)	894	235	0	204	43	1376
Energy Met (MU)	1501	1500	1330	648	80	5058
Hydro Gen (MU)	337	133	155	93	28	746
Wind Gen (MU)	5	76	156	-	-	236
Solar Gen (MU)*	153.43	92.96	134.78	3.52	1.20	386
Energy Shortage (MU)	5.62	9.26	0.00	1.35	0.19	16.42
Maximum Demand Met During the Day (MW) (From NLDC SCADA)	68639	67007	64870	30526	3855	227048
Time Of Maximum Demand Met	20:28	15:48	10:37	23:01	17:55	11:45

B. Frequency Profile (%)

Region	FVI	< 49.7	49.7 - 49.8	49.8 - 49.9	< 49.9	49.9 - 50.05	> 50.05
All India	0.054	0.34	2.26	10.03	12.63	76.11	11.26

C. Power Supply Position in States

Region	States	Max.Demand Met during the day (MW)	Shortage during maximum Demand (MW)	Energy Met (MU)	Drawal Schedule (MU)	OD(+)/UD(-) (MU)	Max OD (MW)	Energy Shortage (MU)
NR	Punjab	13651	0	264.4	132.9	-0.9	656	0.00
	Haryana	10221	0	214.2	160.1	-0.6	248	0.00
	Rajasthan	13941	0	297.5	100.1	5.2	708	4.85
	Delhi	5386	0	111.2	106.4	0.2	427	0.00
	UP	24493	0	464.1	189.7	-1.2	327	0.00
	Uttarakhand	2245	0	49.3	22.0	0.8	166	0.37
	HP	1826	0	37.2	5.2	1.0	112	0.00
	J&K(UT) & Ladakh(UT)	2620	132	53.1	32.8	2.4	427	0.40
	Chandigarh	313	0	6.2	5.9	0.3	58	0.00
WR	Railways_NR ISTS	198	0	4.0	3.7	0.3	58	0.00
	Chhattisgarh	5766	23	130.8	75.3	-0.1	391	0.81
	Gujarat	23268	0	481.7	180.4	-3.8	414	0.00
	MP	11556	93	245.3	115.5	-2.3	560	0.20
	Maharashtra	25491	2056	566.7	171.7	4.3	1082	8.16
	Goa	663	31	14.6	11.1	3.1	128	0.09
	DNHDDPDCL	1328	0	30.7	30.8	-0.1	57	0.00
	AMNSIL	808	0	17.5	7.5	0.2	238	0.00
	BALCO	527	0	12.6	12.3	0.3	62	0.00
SR	Andhra Pradesh	12243	0	242.6	94.9	-1.1	660	0.00
	Telangana	15570	0	299.5	153.8	-1.5	496	0.00
	Karnataka	16050	0	281.7	105.3	2.2	1123	0.00
	Kerala	4229	0	86.8	53.4	0.4	332	0.00
	Tamil Nadu	18515	0	408.0	171.6	-3.0	817	0.00
	Puducherry	516	0	11.2	10.9	-0.3	33	0.00
ER	Bihar	7686	204	159.5	150.1	-2.1	544	1.35
	DVC	3415	0	73.2	-51.0	-2.1	179	0.00
	Jharkhand	2185	0	43.4	33.7	0.5	127	0.00
	Odisha	6001	0	131.3	28.6	-1.4	521	0.00
	West Bengal	11881	0	238.6	99.9	-1.3	321	0.00
	Sikkim	90	0	1.5	1.6	-0.1	13	0.00
	Railways_ER ISTS	15	0	0.1	0.1	0.0	12	0.00
NER	Arunachal Pradesh	181	0	3.6	3.0	0.2	31	0.00
	Assam	2687	0	55.8	47.4	1.2	132	0.11
	Manipur	214	18	3.2	3.2	0.1	37	0.08
	Meghalaya	257	0	4.7	1.2	-0.3	109	0.00
	Mizoram	129	0	2.4	0.6	0.0	15	0.00
	Nagaland	165	0	3.3	2.9	0.0	31	0.00
	Tripura	376	0	7.3	6.3	0.9	83	0.00

D. Transnational Exchanges (MU) - Import(+ve)/Export(-ve)

	Bhutan	Nepal	Bangladesh	Godda -> Bangladesh
Actual (MU)	36.2	17.8	-22.6	-24.9
Day Peak (MW)	1975.0	866.0	-986.0	-1123.1

E. Import/Export by Regions (in MU) - Import(+ve)/Export(-ve); OD(+)/UD(-)

	NR	WR	SR	ER	NER	TOTAL
Schedule(MU)	243.1	-245.5	101.1	-113.2	14.5	0.0
Actual(MU)	241.1	-239.9	98.1	-120.7	15.8	-5.7
O/D/U/D(MU)	-2.0	5.6	-3.1	-7.5	1.3	-5.7

F. Generation Outage(MW)

	NR	WR	SR	ER	NER	TOTAL	% Share
Central Sector	3788	12080	3658	4376	280	24182	49
State Sector	6266	9460	8222	1144	155	25246	51
Total	10054	21540	11880	5520	435	49428	100

G. Sourcewise generation (Gross) (MU)

	NR	WR	SR	ER	NER	All India	% Share
Coal	778	1422	688	730	17	3636	67
Lignite	28	15	64	0	0	108	2
Hydro	337	133	155	93	28	746	14
Nuclear	30	53	71	0	0	154	3
Gas, Naptha & Diesel	10	61	5	0	23	99	2
RES (Wind, Solar, Biomass & Others)	164	171	319	5	1	660	12
Total	1347	1856	1303	828	70	5405	100

Share of RES in total generation (%)	12.15	9.20	24.52	0.64	1.71	12.22
Share of Non-fossil fuel (Hydro,Nuclear and RES) in total generation(%)	39.38	19.25	41.85	11.87	42.34	28.88

H. All India Demand Diversity Factor

Based on Regional Max Demands	1.034
Based on State Max Demands	1.086

I. All India Peak Demand and shortage at Solar and Non-Solar Hour

	Max Demand Met(MW)	Time	Shortage(MW)
Solar hr	227048	11:45	0
Non-Solar hr	214862	19:22	65

Diversity factor = Sum of regional or state maximum demands / All India maximum demand

**Note: All generation MU figures are gross

***Godda (Jharkhand) -> Bangladesh power exchange is through the radial connection (isolated from Indian Grid)

Solar Hours -> 06:00 to 18:00hrs and rest are Non-Solar Hours

*Source: RLDCs for solar connected to ISTS; SLDCs for embedded solar. Limited visibility of embedded solar data.

Executive Director-NLDC

INTER-REGIONAL EXCHANGES

Import=(+ve) /Export =(-ve) for NET (MU)

Date of Reporting: 21-Sep-2024

Sl No	Voltage Level	Line Details	No. of Circuit	Max Import (MW)	Max Export (MW)	Import (MU)	Export (MU)	NET (MU)
Import/Export of ER (With NR)								
1	HVDC	ALIPURDUAR-AGRA	2	0	350	0.0	8.2	-8.2
2	HVDC	PUSAULI B/B	-	0	112	0.0	2.8	-2.8
3	765 kV	GAYA-VARANASI	2	189	552	0.0	3.6	-3.6
4	765 kV	SASARAM-FATEHPUR	1	112	256	0.0	0.9	-0.9
5	765 kV	GAYA-BALIA	1	0	590	0.0	10.5	-10.5
6	400 kV	PUSAULI-VARANASI	1	0	180	0.0	3.5	-3.5
7	400 kV	PUSAULI -ALLAHABAD	1	69	0	0.9	0.0	0.9
8	400 kV	MUZAFFARPUR-GORAKHPUR	2	0	417	0.0	6.1	-6.1
9	400 kV	PATNA-BALIA	2	0	642	0.0	9.8	-9.8
10	400 kV	NAUBATPUR-BALIA	2	0	187	0.0	2.4	-2.4
11	400 kV	BIHARSHARIFF-BALIA	2	124	17	2.2	0.0	2.2
12	400 kV	MOTIHARI-GORAKHPUR	2	0	332	0.0	5.1	-5.1
13	400 kV	BIHARSHARIFF-VARANASI	2	0	0	0.0	0.0	0.0
14	220 kV	SAHUPURI-KARAMNASA	1	0	67	0.0	1.3	-1.3
15	132 kV	NAGAR UNTARI-RIHAND	1	0	0	0.0	0.0	0.0
16	132 kV	GARWAH-RIHAND	1	0	0	0.6	0.0	0.6
17	132 kV	KARMANASA-SAHUPURI	1	0	0	0.0	0.0	0.0
18	132 kV	KARMANASA-CHANDAULI	1	0	0	0.0	0.0	0.0
ER-NR						3.8	54.4	-50.6
Import/Export of ER (With WR)								
1	765 kV	JHARSUGUDA-DHARAMJAIGARH	4	121	909	0.0	10.1	-10.1
2	765 kV	NEW RANCHI-DHARAMJAIGARH	2	1997	46	25.1	0.0	25.1
3	765 kV	JHARSUGUDA-DURG	2	0	872	0.0	13.9	-13.9
4	400 kV	JHARSUGUDA-RAIGARH	4	0	724	0.0	12.0	-12.0
5	400 kV	RANCHI-SIPAT	2	373	190	2.5	0.0	2.5
6	220 kV	BUDHIPADAR-RAIGARH	1	0	182	0.0	2.8	-2.8
7	220 kV	BUDHIPADAR-KORBA	2	0	168	0.0	2.8	-2.8
ER-WR						27.6	41.6	-14.0
Import/Export of ER (With SR)								
1	HVDC	JEYPORE-GAZUWAKA B/B	2	0	730	0.0	17.4	-17.4
2	HVDC	TALCHER-KOLAR BIPOLE	2	0	2	0.0	0.0	0.0
3	765 kV	ANGUL-SRIKAKULAM	2	0	2613	0.0	44.1	-44.1
4	400 kV	TALCHER-I/C	2	1876	0	40.9	0.0	40.9
5	220 kV	BALIMELA-UPPER-SILERRU	1	0	0	0.0	0.0	0.0
ER-SR						0.0	61.5	-61.5
Import/Export of ER (With NER)								
1	400 kV	BINAGURI-BONGAIGAON	2	0	603	0.0	11.4	-11.4
2	400 kV	ALIPURDUAR-BONGAIGAON	2	0	1018	0.0	18.3	-18.3
3	220 kV	ALIPURDUAR-SALAKATI	2	0	196	0.0	3.5	-3.5
ER-NER						0.0	33.2	-33.2
Import/Export of NER (With NR)								
1	HVDC	BISWANATH CHARIALI-AGRA	2	0	853	0.0	20.4	-20.4
NER-NR						0.0	20.4	-20.4
Import/Export of WR (With NR)								
1	HVDC	CHAMPA-KURUKSHETRA	2	0	2014	0.0	36.1	-36.1
2	HVDC	VINDHYACHAL B/B	-	48	58	0.0	1.2	-1.2
3	HVDC	MUNDRA-MOHINDERGARH	2	0	1903	0.0	35.4	-35.4
4	765 kV	GWALIOR-AGRA	2	0	1470	0.2	17.9	-17.7
5	765 kV	GWALIOR-PHAGI	2	360	1787	0.5	22.8	-22.3
6	765 kV	JABALPUR-ORAI	2	0	1110	0.0	26.7	-26.7
7	765 kV	GWALIOR-ORAI	1	698	0	10.5	0.0	10.5
8	765 kV	SATNA-ORAI	1	0	881	0.0	14.8	-14.8
9	765 kV	BANASKANTHA-CHITORGARH	2	1795	196	21.7	0.1	21.6
10	765 kV	VINDHYACHAL-VARANASI	2	0	3456	0.0	42.8	-42.8
11	400 kV	ZERDA-KANKROLI	1	374	79	3.9	0.1	3.8
12	400 kV	ZERDA -BHINMAL	1	42	221	0.0	2.0	-2.0
13	400 kV	VINDHYACHAL -RIHAND	1	0	1	0.0	0.0	0.0
14	400 kV	RAPP-SHUJALPUR	2	341	469	1.7	4.0	-2.4
15	400 kV	NEEMUCH-Chittorgarh	2	193	546	0.5	5.4	-4.9
16	220 kV	BHANPURA-RANPUR	1	0	79	0.0	1.0	-1.0
17	220 kV	BHANPURA-MORAK	1	0	30	0.0	2.2	-2.2
18	220 kV	MEHGAON-AURAIYA	1	90	4	0.8	0.0	0.8
19	220 kV	MALANPUR-AURAIYA	1	67	13	0.5	0.0	0.4
20	132 kV	GWALIOR-SAWAI MADHOPUR	1	0	0	0.0	0.0	0.0
21	132 kV	RAJGHAT-LALITPUR	2	0	0	0.0	0.0	0.0
WR-NR						40.1	212.5	-172.4
Import/Export of WR (With SR)								
1	HVDC	BHADRAWATI B/B	-	0	1007	0.0	22.6	-22.6
2	HVDC	RAIGARH-PUGALUR	2	0	2003	0.0	29.0	-29.0
3	765 kV	SOLAPUR-RAICHUR	2	1144	1127	6.7	3.4	3.3
4	765 kV	WARDHA-NIZAMABAD	2	116	2494	0.1	31.0	-30.9
5	765 kV	WARORA-WARANGAL(NEW)	2	69	2309	0.0	28.6	-28.5
6	400 kV	KOLHAPUR-KUDGI	2	1405	0	24.6	0.0	24.6
7	220 kV	KOLHAPUR-CHIKODI	2	0	0	0.0	0.0	0.0
8	220 kV	PONDA-AMBEWADI	1	81	0	0.0	0.0	0.0
9	220 kV	XELDEM-AMBEWADI	1	0	111	2.3	0.0	2.3
WR-SR						33.7	114.6	-80.9

INTERNATIONAL EXCHANGES					Import(+ve)/Export(-ve)		
State	Region	Line Name	Max (MW)	Min (MW)	Avg (MW)	Energy Exchange (MU)	
BHUTAN	ER	400kV MANGDECHHU-ALIPURDUAR 1,2&3 i.e. ALIPURDUAR RECEIPT (from MANGDECHU HEP 4*180MW)	741	551	608	14.58	
	ER	400kV TALA-BINAGURI 1,2,4 (& 400kV MALBASE - BINAGURI) i.e. BINAGURI RECEIPT (from TALA HEP 6*170MW)	853	530	701	16.82	
	ER	220kV CHUKHA-BIRPARA 1&2 (& 220kV MALBASE - BIRPARA) i.e. BIRPARA RECEIPT (from CHUKHA HEP 4*84MW)	242	55	101	2.42	
	NER	132kV GELEPHU-SALAKATI	40	0	33	0.79	
	NER	132kV MOTANGA-RANGIA	74	0	64	1.54	
NEPAL	NR	NEPAL IMPORT (FROM UP)	0	0	0	0.00	
	NR	132kV MAHENDRANAGAR-TANAKPUR(NHPC)	64	0	44	1.06	
	ER	NEPAL IMPORT (FROM BIHAR)	64	38	43	1.04	
	ER	400kV DHALKEBAR-MUZAFFARPUR 1&2	738	552	653	15.66	
BANGLADESH	ER	BHERAMARA B/B HVDC (B'DESH)	-900	-746	-874	-20.97	
	ER (Isolated from Indian Grid)	400kV GODDA_TPS-RAHANPUR (B'DESH) D/C	-1123	-1002	-1038	-24.92	
	NER	132kV COMILLA-SURAJMANI NAGAR 1&2	-86	0	-66	-1.58	

Export From India (in MU)

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Nepal	0.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.10
Bangladesh	21.69	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.69
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Export	21.79	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.79

Import by India(in MU)

Country	GNA (ISGA/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	34.88	0.00	1.65	0.00	0.00	0.06	0.00	0.00	36.59
Nepal	7.36	0.00	11.26	0.00	0.00	0.00	0.00	0.00	18.62
Bangladesh	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Import	42.24	0.00	12.91	0.00	0.00	0.06	0.00	0.00	55.21

Net from India(in MU)

-ve : Export / +ve : Import

Country	GNA (ISGS/PPA)	T-GNA							TOTAL
		BILATERAL TOTAL	COLLECTIVE						
			IDAM			RTM			
			IEX	PXIL	HPX	IEX	PXIL	HPX	
Bhutan	34.88	0.00	1.65	0.00	0.00	0.06	0.00	0.00	36.59
Nepal	7.26	0.00	11.26	0.00	0.00	0.00	0.00	0.00	18.52
Bangladesh	-21.69	0.00	0.00	0.00	0.00	0.00	0.00	0.00	-21.69
Myanmar	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total Net	20.45	0.00	12.91	0.00	0.00	0.06	0.00	0.00	33.42